

Programming Fundamentals – COC1071

Lab Plan – 09



Learning Objective:

- *Default Arguments*
- *Function Overloading*

- 1) Write a function **greet** that takes a name as a parameter and has a default argument for the greeting (e.g., "Hello"). If no greeting is provided, it should use the default.
- 2) Write a function **calculatePrice** that takes the price of an item, quantity, and a tax rate. Set a default tax rate (e.g., 5%) and quantity (e.g., 1) if not provided.
- 3) Write a function **displayDate** that takes day, month, and year parameters, using default values for month and year (e.g., current month and year). Display the full date.
- 4) Write a function **printMultiplicationTable** that takes a number and a limit (default 10) to generate its multiplication table up to the specified limit.
- 5) Write a function **drawRectangle** that takes width and height, using a default height of 5 and width of 10. Print the rectangle with * symbols.
- 6) Create a function **sumUpTo** that takes an integer n and an optional start (default 1) to calculate the sum of numbers from start to n.
- 7) Overload a function **calculateArea** to calculate the area of a square, rectangle, and circle. Use one parameter for the square, two for the rectangle, and one for the circle (with $\pi * r * r$).
- 8) Overload a function **findMax** to return the maximum of two integers, two doubles, and three integers.
- 9) Overload a **power** function to calculate the power of an integer raised to another integer, a double raised to an integer, and an integer raised to a double.
- 10) Write an overloaded **swap** function to swap values of different data types, such as two integers, two doubles, and two characters.
- 11) Overload a function **isEven** that takes an integer, a double, and a character, returning true if the integer or double is even and if the ASCII code of the character is even.
- 12) Overload a function **display** to print names in different formats, such as firstName lastName, lastName, firstName, and firstName initial lastName.
- 13) Write an overloaded **average** function that calculates the average of two integers, three integers, and two double values.